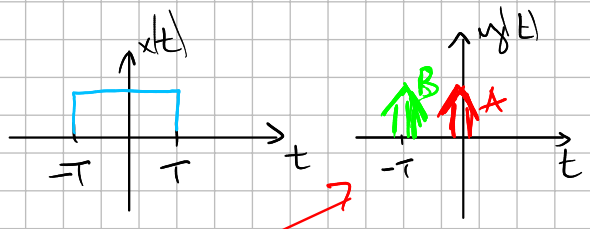
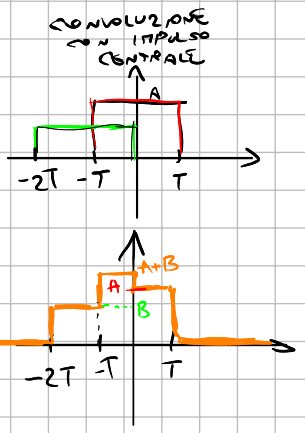
TRACCIARE LA CONVOLUZIONE TRA I SEGNALI $x(t)$ e $y(t)$

$$x(t) = \Pi\left(\frac{t}{2T}\right)$$

$$y(t) = A\,\delta(t) + B\,\delta(t+T) \qquad A > B$$





DETERMINARE LA POTENZA MEDIA DEL SEGNALE

$$s(t) = \cos(2\pi f_0 t)$$

→ NEL TEMPO

→ IN FREQUENZA

con $T_0 = \frac{1}{f_0}$

$$P_s = \lim_{T\to\infty} \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} s^2(t)\,dt = \frac{1}{T_0}\int_{-\frac{T}{2}}^{\frac{T}{2}} s^2(t)\,dt$$

$$= \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \cos^2(2\pi f_0 t)\,dt = \int_{-\frac{T}{2}}^{\frac{T}{2}} \frac{1+\cos(4\pi f_0 t)}{2}\,dt = \frac{1}{T_0}\left[\int_{-\frac{T}{2}}^{\frac{T}{2}} \frac{1}{2}\,dt + \right.$$

$$\left. + \int_{-\frac{T}{2}}^{\frac{T}{2}} \frac{\cos(4\pi f_0 t)}{2}\,dt \right]$$

TUTTI TERMINI NULLI → NESSUN CONTRIBUTO

$$= \frac{1}{T_0}\left[\frac{1}{2}t\right]_{-\frac{T}{2}}^{\frac{T}{2}} = \frac{1}{T_0}\left(\frac{1}{2}\frac{T}{2} + \frac{1}{2}\frac{T}{2}\right)$$

$$= \frac{1}{T_0}\cdot\frac{T}{2} = \frac{1}{2}\,W$$

$$X_k = \frac{2}{T_0}\int_0^{\frac{T_0}{2}} x(t)\cos(2k\pi f_0 t)\,dt$$

$$= \frac{2}{T_0}\int_0^{\frac{T_0}{2}} \cos(2\pi f_0 t)\cdot\cos(2k\pi f_0 t)\,dt$$

UNICI COEFF BUONI

$k = \pm 1 \Rightarrow \omega^2$

$k = 0 \Rightarrow X_0 = \phi$

$\begin{cases} k > 2 \\ k < 2 \end{cases} \Rightarrow X_k = \phi$

SINUSOIDE MISURATA IN SERIE SOLO $k \pm 1$ POTEVA ANDARE BENE

$$X_1 = \frac{2}{T_0}\int_0^{\frac{T_0}{2}} \cos^2(2\pi f_0 t)\,dt = \frac{1}{2}$$

$$X_{-1} = \frac{1}{2}$$

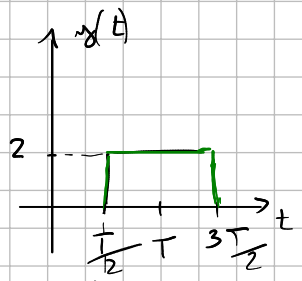
PARSEVAL

$$P_s = X_1^2 + X_{-1}^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} = 0{,}5\,W$$

ESEGUIRE LE CONVOLUZ DEI SEGUENTI SEGNALI

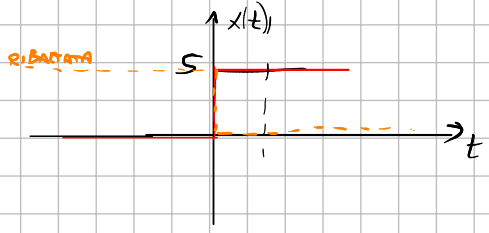
$$x(t) = 5\ w(t)$$

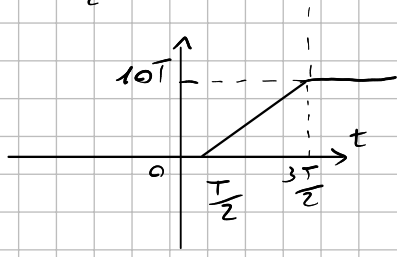
$$y(t) = 2\,\Pi\left(\frac{t - \frac{T}{2}}{T}\right)$$



→ ESEC. X VIA GRAFICA

→ SE x_t RISP IMP, CHE TIPO DI SIST. RAPPRESENTA

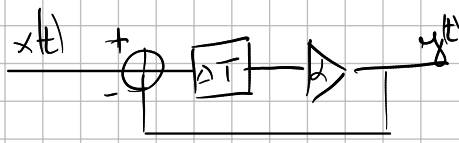




UN SISTEMA CHE RISPONDE CON UN $h(t) = 5\,w(t)$
C'È UNA AMPLIFICAZIONE E UNA INTEGRAZIONE



DEFINIRE UNA RISP. IN FREQ. DEL SISTEMA:



BLOCCO IN RETROAZIONE → DIROTTAMENTE I BLOCCHI NO FORMER

FORMULA PER SIS CON RETROAZIONE NEGATIVA

$$H_{eq}(f) = \frac{H_1(f)}{1 + H_1(f)\,H_2(f)}$$

$$\begin{cases} H_1(f) = \alpha\, e^{-j\,2\pi f \Delta T} \\ H_2(f) = 1 \end{cases}$$

$$H_{eq}(f) = \frac{\alpha e^{-j\,2\pi f \Delta T}}{1 + \alpha\, e^{-j\,2\pi f \Delta T}}$$

CALCOLARE LA DENSITÀ SPETTRALE DI ENERGIA E L'ENERGIA TOTALE DI $x(t)$ t.c.

$$|X(f)| = A\,\Pi\left(\frac{f}{\omega}\right) + B\,\Pi\left(\frac{f}{3\omega}\right)$$

$$|X(t)|^2 = A^2\,\Pi\left(\frac{f}{\omega}\right) + B^2\,\Pi\left(\frac{f}{3\omega}\right) + 2AB\,\Pi\left(\frac{f}{\omega}\right)\cdot\Pi\left(\frac{f}{3\omega}\right)$$

DENS SPETTRALE CORRISPONDENTE — DENS SPETTR CORRISPOND — ENERGIA MUTUA FRA I DUE SEGNALI

$$\frac{\text{Joule}}{\text{Hz}}$$

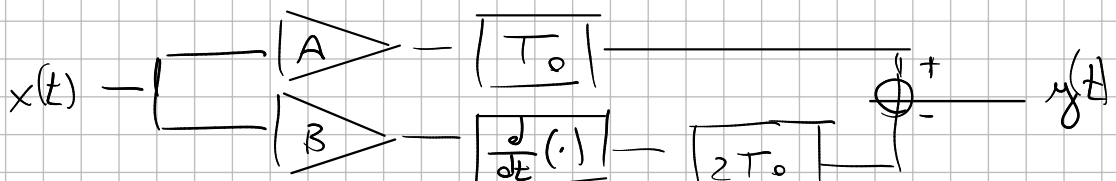
RAYLEIGH

$$E_x = A^2 W + 3B^2 W + 2AB W \quad \text{Joule}$$

RICAVARE LO SCHEMA A BLOCCHI DI UN SISTEMA LTI

$h(t) = A\,\delta(t-T_0) - B\,\delta'(t-2T_0)$

- CALCOLARE LA RISP. IN FREQ $H(f)$



$H(f) = A\,e^{-j2\pi f T_0} - B\,j2\pi f\,e^{-j4\pi f T_0}$

$X(t) = A\,\mathrm{sinc}^2(2\cdot 10^4 t)$

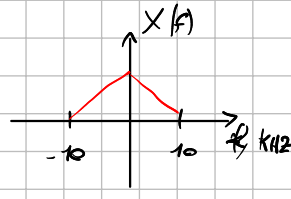
$x(t) \rightarrow (X) \xrightarrow{\cos 2\pi 10^6 t} \boxed{H(f)} \rightarrow y(t)$

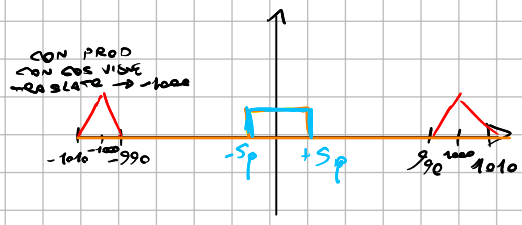
$H(f) = \Pi\left(\frac{f}{10^5}\right)$

CALCOLARE $Y(f)$

RAG → SARÀ UNA FUNZ TRIANGOLO IN FREQ CHE SI ANNULLA IN $2\cdot 10^4$

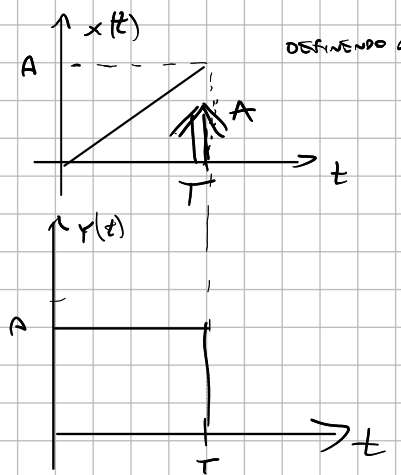
$X(f) = \frac{A}{2\cdot 10^4}\,\Lambda\left(\frac{f}{2\cdot 10^4}\right)$ →



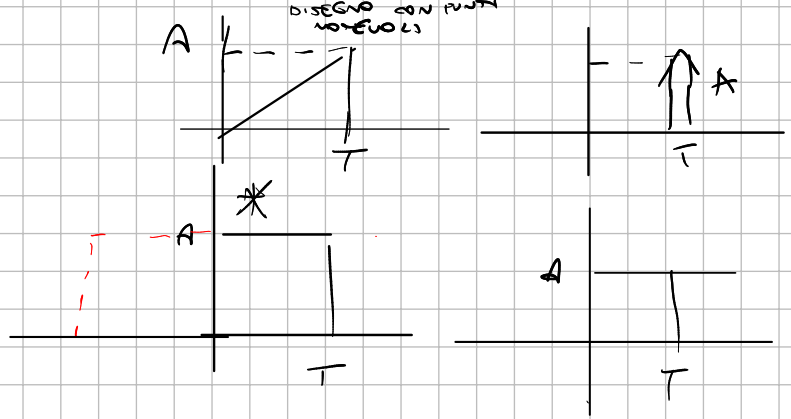


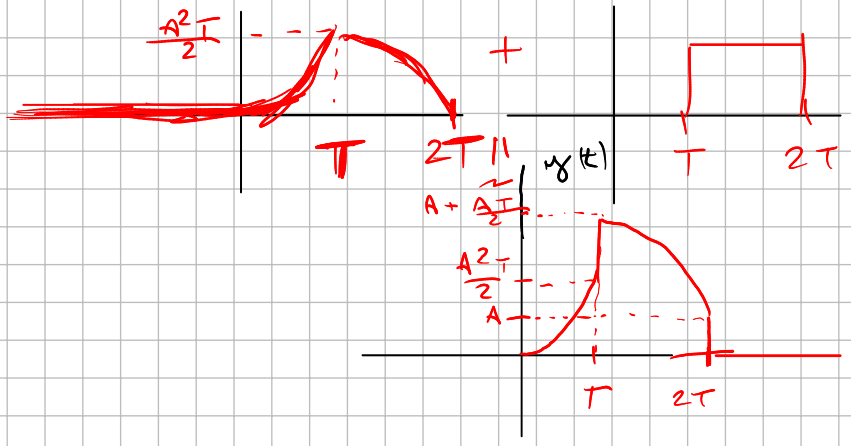
$Y(f) = 0 \rightarrow y(t) = 0$

CONVOLUZ. DEI SEGNALI IN FIGURA



DEFINENDO ANDAMENTO QUALITATIVO → NB FORUNE DISEGNO CON PUNTI NOTEVOLI




$\frac{A^2T}{2}$
T
2T
+
T
2T
$y(t)$
$A+\frac{A^2T}{2}$
$\frac{A^2T}{2}$
A
T
2T